

AE460B Spacecraft Design

TM-04: Onboard Data Handling (OBDH) Design

Destiny

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In this paper, a design overview of the Onboard Data Handling (OBDH) subsystem of the Solar Gravitational Lens (SGL) spacecraft is developed. In order to meet the 650 AU in 25 year requirement, along with image acquisition and down-link at that distance, a necessary 68 W, 7.3 kg system is needed to handle all data on the craft. With this, it will be shown that 4 TB of memory along with 100 MHz CPU is required to handle payload data and processing. Throughout the paper, it shown these specs are achievable with current or near-future technologies.

Nomenclature

<i>OBDH</i>	= Onboard Data Handling System
<i>SGL</i>	= Solar Gravitational Lens Mission
<i>ADCS</i>	= Attitude Determination and Control System
<i>COM</i>	= Communications Subsystem
<i>EPS</i>	= Electrical Power Subsystem
<i>RTOS</i>	= Real-Time Operating System
<i>FDIR</i>	= Failure Detection, Isolation, and Recovery
<i>TC</i>	= Telecommand
<i>TM</i>	= Telemetry
<i>OBC</i>	= Onboard Computer
<i>DSAC</i>	= Deep Space Atomic Clock
<i>RTG</i>	= Radioisotope Thermoelectric Generator
<i>PCU</i>	= Power Conditioning Unit
<i>FPGA</i>	= Field Programmable Gate Array
<i>MLI</i>	= Multi-Layer Insulation
P_{cpu}	= Required CPU clock speed
M_{RAM}	= Volatile memory (RAM)
M_{NVM}	= Non-volatile memory
M_{MM}	= Mass memory capacity
T_{HK}	= Housekeeping sampling window
N_{TC}	= Number of stored time-tagged commands
R_{bus}	= Aggregate data rate
ρ_t	= Radiation tolerance
$V_{HK,raw/day}$	= Raw housekeeping data per day
$V_{pl,raw/day}$	= Raw payload data per day
$V_{HK,eff/day}$	= Effective housekeeping data per day
$V_{pl,eff/day}$	= Effective payload data per day
f_{oh}	= Packet overhead factor
f_{retx}	= Retransmission factor
m	= Memory/CPU margin factor
CR	= Compression ratio

N_{pre}	=	Pre-science mission duration (days)
N_{mid}	=	Mid-science mission duration (days)
F	=	Growth factor $(1 + f_{oh})(1 + f_{retx})(1 + m)$
$B_{TM TC}$	=	TM/TC processing throughput (bytes/s)
$C_{TM TC}$	=	CPU cycles per second for TM/TC
$U_{TM TC}$	=	CPU utilization for TM/TC
k_{proto}	=	Protocol cost (cycles/byte)
t_{ADCS}	=	ADCS computation time per cycle
f_{ADCS}	=	ADCS update frequency
U_{ADCS}	=	CPU utilization for ADCS
k_{comp}	=	Compression cost (cycles/byte)
C_{comp}	=	Payload compression CPU load
U_{comp}	=	CPU utilization for compression
U_{bus}	=	Bus utilization fraction
R_i	=	Data rate of subsystem i
$R_{bus,max}$	=	Maximum bus throughput

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I. Mission Context and OBDH Requirements

A. Mission Scenario

The Solar Gravitational Lense (SGL) mission's goal is to reach 650 AU from the Sun within 25 years. Through this process, it will reach within 0.14 AU from the Sun where it is subject to extremely high temperature and large amounts of solar radiation. Additionally, the spacecraft will spend a majority of its time outside the heliosphere, where it will no longer be shielded from background radiation. During this period of cruise, being able to monitor the health and position of the spacecraft will be crucial in order to make sure it is reaching the desired location. The ability to give commands will also be necessary in case there is any deviations or errors that must be fixed; however, positional correction will be done automatically by the spacecraft during its cruise.

Once the craft reaches the desired location, the focus shifts to imaging the pictures from the Sun, processing it in house, and then sending the pictures through a downlink antenna to Earth. Additionally, the spacecraft will have to enter into a controlled spiral motion to adjust for both the wobbling of the Sun and the orbit of the planet that is being imaged. These two processes require an extreme amount of data to be sent, along with all housekeeping telemetry such as thermal conditions or fuel amount.

Thankfully, due to the fact that the imaging position is out of plane of the orbit of Earth around the Sun, communication with Earth will be available at all times, reducing necessary storage on the craft and the max bit-rate required.

B. OBDH Core Functions

The On Board Data Handling (OBDH) system's main purpose is to collect all of this necessary data, process it, and determine where it needs to go. For this specific mission, the OBDH will have to:

- process and transmit all housekeeping data, which includes electrical output from radioisotope thermal-electric generators and deployment of the solar sail.
- determine the position of spacecraft using star tracker data
- calculate thrust direction and magnitude needed to obtain necessary position during cruise and imaging
- store, process, and transmit image data during imaging mode
- work with volatile memory required for image processing
- receive, process, and execute any command sent from the ground station
- keep track of time, which is necessary for automation
- look for and disable any faulty hardware caused by radiation, while relaying that to the ground station

C. Key Requirements

In order to do everything stated in the previous section, the OBDH will have to have the follow minimum specifications.

For the processing power, an approximate 36 MHz is required. This number came from taking the maximum bit rate, which is about 12 kbps, and estimating that approximately 300 operations happens per bit. This gives 36 million operations per second.

For the RAM, technically a minimum of around 100 MB is actually required to do the image processing at the necessary speed, but in order to account for any radiation and housekeeping 16 GB is chosen.

For non-volatile memory, an approximation of 1 GB was chosen as the necessary space to hold the downloaded information for the controls for all the subsystems while also accounting for any radiation that may damage the system. 32 GB of mass memory capacity is chosen to hold the 8 hours of sampling window from the imaging.

The time-tagged command storage is quite low since most commands will happen nearly as they are sent as the spacecraft always has a view of Earth during imaging. However, during movement in the solar system, some time-tagged events need to be allowed as we will not have full vision, so 10 is chosen arbitrarily.

The central data bus type chose is SP0-S. This will be explained more later, but the main reason is because it is lightweight and has the high radiation tolerance required for the mission.

The amount of radiation tolerance was determined by taking the radiation in deep space past the heliosphere (which is at maximum 70 rad/yr) and multiplying it by 25 years to get the maximum the craft should theoretical experience which is 2 krad. There is also some radiation from the Sun, which is why the limit was raised to 4 krad.

Characteristic	Requirement
P_{cpu}	5 MHz
M_{RAM}	16 GB
M_{NVM}	1 GB
M_{MM}	3.845 TB
T_{HK}	8 hours
N_{TC}	10 commands/sequences
Central Bus Type	SP0-S
R_{bus}	20 Kbps
ρ_t	4 kRad
link/interface to TT&C	SpaceWire
T Range	-20 to 50 °C

Table 2 List of Different Requirements for SGL Mission's OBDH

II. OBDH Architecture Selection

A. Architecture Concept

The main idea conflict the spacecraft has with the environment will be dealing with long term effects of radiation. In order to combat this, there will be multiple buses. For housekeeping, the SP0-S will be used. SP0-S was chosen because of it is design to last a long time in high radiation environments. Also, it is light weight and low power for how fast it can process. The communication connections used on board is SpaceWire. It has low density, highly radiation resistant cables. These properties are extremely helpful for a mission where minimizing mass is more important than usual. Afterwards, the data gets sent to the encoder. For up-link, the process will be in reverse with a decoder instead.

Below is a table of the different subsystems and characteristics of their processing unit. Note that when it says <2kbps it means that the main data of the subsystem is housekeeping and not enough is known to get a estimate. However, all of these will be lumped into one big housekeeping estimate later on in the paper.

Subsystem	Interface	Data rate (kbps)	Update frequency (MHz)
Electrical	SP0-S	<2	400
Thermal	SP0-S	<2	400
Propulsion	SP0-S	<2	400
ADCS	SP0-S	1.2	400
COM	SP0-S	8	400
Payload	NVIDIA Jetson Nano (Shielded)	5	921.6

Table 3 Subsystems and their Processing Method

Note that the payload has a different interface. The NVIDIA Jetson Nano is chosen for it's incredible power optimization for complex processes, minimizing the data rate and power consumption for the image processing.

B. Functional Partitioning

For this section, the functions of the spacecraft are laid out, changing with the different states the mission is in. Throughout the process, changes are made to the order by attempting to minimize the things need to be run at once in order to reduce necessary RAM.

1. Launch

At launch, the main important computer systems are just the housekeeping for the health of the spacecraft. In this case, the processing power will be dedicated to the housekeeping.

2. Nominal

These are cases where the spacecraft is traveling between two points of interest. In this case, focus will be on the star tracking. The partitioning will cycle between collection of data and the sending of data in order to minimize power and RAM usage.

3. Jupiter

In Jupiter, communication will be cut and the craft will have to handle the propulsion itself. In this case, all processing power will go into exactly determining the craft's orientation and position.

4. Sun

During this phase, similar to the Jupiter phase, most of the processing power will be sent controlling the orientation and position of the spacecraft. Additionally, some initial RAM will be given to the deployment and monitoring of its solar sail before perihelion, while also relaying the progress to the ground station.

During the solar Oberth, the spacecraft will automatically aim the solar sail and heat shield towards the Sun, technology that was demonstrated during the Parker Solar Probe perihelion. Also during this phase communication will be cut off.

5. Imaging

During imaging, the satellite will enter into a spiral whose location will be determined using the star tracker. Additionally, the spacecraft will be pointing towards the Sun and taking images. This is where the most processing power will be used, since along with positioning, the images will be processed onboard. However, editing (such as coloring or fixing artifacts) of the image will be done on the ground.

If the spacecraft is normal, no partitioning should be required. However, if there is radiation damage from the cruise, the spacecraft will then focus mainly on propulsion, thermal, and imaging and relay housekeeping information every hour.

6. Post Mission

The imaging of the satellite will be turned off. Any auxiliary scientific missions (such as the missions on the Voyager probes) installed on the spacecraft will be the only things running along with housekeeping data sent once every day.

As power decreases, partitioning will become more important. The housekeeping of propulsion will decrease in importance, then the extra processors will be turned off keeping only necessary ones. At the end of the mission, the scientific instruments will be shut down based off their importance.

III. Modes, States, and Timeline Integration

A. Modes and States

The OBDH will have a variety of modes, states, and transitions, each designed for the spacecraft's major mission phases. Modes will impact task execution priorities, on-board health systems such as power allocation and telemetry, and autonomous behavior. Transitions between OBDH states are triggered by time-tagged commands at pre-planned moments, event detections, or Failure Detection Isolation, and Recovery (FDIR) system fault conditions. Each operational mode is described below.

Mode 1: Pre-Launch and Initialization

This mode activates when the spacecraft is powered on before launch. Its core purpose is to perform a system health checkout that will detect memory or software issues, telemetry streams, time reference, and connection with onboard interfaces. The results of this self-test will inform whether the spacecraft will continue to launch or remain in initialization.

Mode 2: Launch and Nominalization

This mode is activated during launch and will continue until the spacecraft is in a nominal configuration. The purpose of this mode is to get the spacecraft into an operational state. The OBDH will focus on ADCS stabilization and the establishment of a communication link with ground stations. Mechanical components for a stored configuration will be triggered to move the spacecraft into its cruise configuration. Once these conditions have been met, the next mode will trigger.

Mode 3: Cruise

This mode will be active during all steps of the 25+ year cruise to 650 AU. The focus of this mode will be the execution of trajectory maneuvers and low-rate communications maintenance. The OBDH will use time-tagged commands to perform precalculated trajectory maneuvers that include the Earth-Jupiter transfer, solar infall, and hyperbolic escape. During these modes, the OBDH will be performing nominal surveillance, memory duplication, RAM checks, and other rigorous steps to protect against radiation-related issues, which can persist due to the long mission duration. After the perihelion maneuver, this mode will also jettison the thermal heat shield and allow the spacecraft to enter its science gathering configuration.

Mode 4: Science Gathering

Upon reaching 650 AU, Science Gathering Mode will be triggered with a primary goal of supporting the continuous image acquisition, digitization, and preparation of science data. The payload subsystem will continually gather time-stamped exposures of the SGL. This data will be processed and stored in a state ready for scheduled downlink windows. Science Gathering Mode will be active from 650 to 950 AU.

Mode 5: High-Rate Downlink

During scheduled downlink windows, the OBDH will enter high-rate downlink mode with the primary goal of sending packets of data directly to Earth. The optical laser communication system will begin a continuous stream of packeted science data while the OBDH monitors ADCS, electrical, and thermal systems to throttle the data rate if needed. This mode will be the highest power consumption configuration outside of propulsion events. Once transmission windows are over, the OBDH will re-enter Science Gathering Mode.

Mode 6: End-of-life

After 950 AU, the final OBDH mode will be triggered, which will systematically put the spacecraft into a graveyard state with all essential and supporting functions turned off as the mission comes to an end.

Mode 7: Safe Mode Safe mode is an autonomous state that will be activated to protect the spacecraft during anomalies. This mode can be triggered by the FDIR or anomalous telemetry events. During this mode, high-rate downlink is disabled, the processor load is reduced, and thermal systems are prioritized. This mode will continue until intervention from ground support or if autonomous solutions can return telemetry to nominal values.

The State Transition Diagram shows how Mode 7 is used to procedurally test errors and safeguard the system as it moves linearly through the planned mission phases. Additionally, the OBDH confirms that Mode 1 tests are cleared before allowing a Launch configuration.

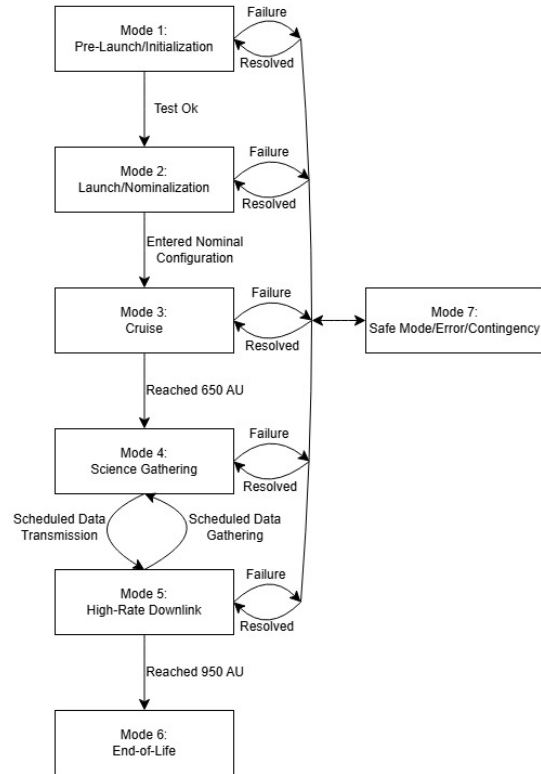


Fig. 1 OBDH Operational Mode State Transition Diagram.

B. Time Distribution

The OBDH will maintain an expected high accuracy <1 millisecond time difference with Earth-time to keep time-tagged commands happening at precise moments. The OBDH will use two sources for time reference. The primary source will be an onboard oscillator; the only current high Technology Readiness Level (TRL) option is the Deep Space

Atomic Clock (DSAC), a bulky yet extremely precise clock designed for deep space missions. The DSAC would keep the time synchronization jitter within 10 microseconds between synchronized subsystems, which would enable the high level of precision needed between the ADCS and optical communications. The secondary source of time reference will be periodic uplink telecommands that can realign any slight clock drifts over the decades of operation.

IV. Sizing Methodology

A. Data Rates and Data Volume

Because the SGL is holding a single target in space and is away from the influences of any gravity, the station-keeping required is minimal. Despite this, there will need to be constant minor maneuvers throughout its journey in order to follow the stellar wobbles that occur from the target planet moving in relation to the SGL and our sun [1]. Part of the uplink for this mission will be to analyze the current parameters of the spacecraft and plan 1-2 daily burns to keep the spacecraft within proper visual holding of the target planet. These maneuver uploads (which may only be performed once every 1-2 weeks, along with minor housekeeping uploads such as clock adjustments, and any critical adjustments to hardware) will mean that the required Uplink will be on the order of 15 kilobytes/s. This extremely minimal rate will be held for nearly the entirety of the spacecraft's lifetime. The downlink for the spacecraft will be roughly the same. The two primary transmissions from the spacecraft will be between position/attitude TLM and payload image results. Because the payload will not begin functioning until halfway through its lifetime, the standard downlink before science will likely be >5 kilobytes/s. Once the science has begun, image processing will be done on board, making the downlink far lighter.

As mentioned, the Payload science will not begin until the SGL has reached a distance of ~650 AU, which is the distance at which the Einstein ring becomes visible, and image collection can begin. During the data collection portion, light will be continuously collected from the Einstein ring, which is then processed and stored on the On Board Computer (OBC). Because the nature of the image processing is layering new data on top of an already processed image, there will be no peak in the data rates, only in the storage of the collected data. The data is retained onboard until reconstructed images are verified and scheduled for transmission, after which raw measurements may be selectively discarded to free storage capacity. The storage for the payload, as well as any data in the OBDH, is set at 1 Tb. Storage of this size allows for nearly all data collected throughout the spacecraft timeline to be stored for future downlink without the danger of overpopulating the total storage capacity. With 1 Tb, the OBDH should never be held above 20% the total storage, ensuring long-term survival. The only event-based data burst comes from the initial detection of the Einstein ring, and would not have a substantial burst in data use due to the already processing speed required.

B. Mass Memory Sizing

The memory required for the mission is not substantial, which is a requirement due to the difficulty of maintaining electronics in near absolute zero, where the only heat comes from radiation. For the bus, the time it is required to be operating is roughly 39.43 years (14,400 days). For the payload, which activates at 650 AU (given a speed of 123.5 km/s), it will only be operating for 9.6 years (3,500 days). Retransmissions come from extra data that failed to downlink—either through errors, dropped frames, or weather—and need to be resent once confirmed to be lost. Due to the extreme accuracy we are expecting, and the chance for weather at any of the four ground stations, and a low Signal to Noise Ratio (SNR), we assume a 20% retransmission factor (f_{retx}). Packet overhead, which is the wrapping around the TLM to organize it with timestamps, sync up words, and checksum verification, is always prevalent in spacecraft telemetry. In our case, because there is a constant stream of larger packets sending data, we can assume a smaller factor of data rates than normally influences other spacecraft. Our SGL will work with a factor of 10% packet overhead (f_{oh}).

There are a number of uncertainties to consider that may change as the mission progresses, especially since most of the technology utilized in the SGL is not field-tested. Between the compression ratio, CPU and bus sizing adjustments, final housekeeping size, and many other factors, we adjust for these uncertainties with a Memory margin and CPU margin of 40% (m). Because the image processing is done on board the spacecraft, and the entire purpose of the mission is to deliver a high-quality image of another planet, we are allowing for a 2:1 Compression Ratio (CR), which will deliver the best image we possibly can, given the constraints. As mentioned, there are two stages to the memory requirements on the mission. The first 10,900 days are the pre-science stage (N_{pre}), where only the bus contributes to the mass memory, and the mid-science covers the remaining 3,500 days (N_{mid}). The equations for the required Mass Memory at each stage is found to be:

$$M_{MM,req,pre} \approx V_{HK,eff/day} \times N_{pre} \quad (1)$$

$$M_{MM,req,mid} \approx M_{MM,req,pre} + (V_{HK,eff/day} + V_{pl,eff/day}) \times N_{mid} \quad (2)$$

The data volumes for the Bus and the Payload are given as follows:

$$V_{HK,eff/day} = V_{HK,raw/day} \times (1 + f_{oh}) \times (1 + f_{retx}) \times (1 + m) \quad (3)$$

$$V_{pl,eff/day} = V_{pl,raw/day} \times (1 + f_{oh}) \times \frac{1}{CR} \times (1 + f_{retx}) \times (1 + m) \quad (4)$$

Using the factors assumed based on retransmissions, packet overhead, compression ratios, and memory margin, we have all of the needed factors to solve for the mass memory. With the pre-established 20 Kb/s raw downlink speed, roughly 15 Kb/s of that dedicated to the bus, and the rest towards the payload, we can solve both the pre-science and mid-science mass memory to be 2.797 TB and 1.048 TB, respectively. This means that by the end of the mission, the total mass memory would be 3.845 TB. This is clearly far above the designated 1 TB of storage space, which is why we will have system in place for the housekeeping to filter out older data as new data comes in, which will keep the storage at no higher than 40% total storage, preserving its lifespan. The inclusion of each of these factors found above can be included into a single growth factor (F), which can be used for future calculations:

$$F = (1 + f_{oh})(1 + f_{retx})(1 + m) = 1.584 \quad (5)$$

C. Compute Sizing

For each part of the spacecraft, there may be a number of things that contributes to the CPU work, and therefore distributing available CPU resources becomes an important task to avoid overloading it. This is especially the case in a spacecraft where data resources are so limited. For uplink processing, receiving Telecommands (TC), validating, processing, and rerouting to their intended subsystems contributes to CPU work. Similarly, for downlink, building TLM, packetizing, timestamping, and validating are all to be considered for the CPU. Calculating the TM/TC processing for the CPU is a matter of converting the 504 bits/s for all downlink to bytes, and adjusting with the growth factor found in the previous subsection. From there we further adjust with the software cost (k_{proto}), which we assume to be 2,000 cycles/byte. Once the throughput is calculated, we can get the throughput for the 100 MHz CPU by dividing the throughput by the total power, giving the percent that TM/TC is taking up.

$$B_{TM/TC} = \frac{15000 \cdot F}{8} = 2970 \text{ bytes/s} \quad (6)$$

$$C_{TM/TC} = B_{TM/TC} \cdot k_{proto} = 5940000 \text{ cycles/s} \quad (7)$$

$$U_{TM/TC} = \frac{C_{TM/TC}}{P_{CPU}} = 5.944\% \quad (8)$$

For the Attitude and navigation support, there is also a very small amount of data being reserved for the CPU, and a very similar method of calculation for these figures. For ADCS, we define the support computations to run at a clock time of 5 ms (t_{adcs}), and a Period of 10 Hz (f_{adcs}). With these two times, the CPU time per second for the ADCS subsystem is given as:

$$U_{adcs} = f_{adcs} \cdot t_{adcs} = 5\% \quad (9)$$

This means that 5% of the CPU is taken up by the ADCS, and with the established 100 MHz of power to the CPU, we can find that the ADCS runs at 5,000,000 cycles/s. Running at 10 Hz, each cycle performs quaternion propagation, and processes the sensor input. The 5 ms CPU timing comes from modern star tracker speeds, and will be adequate for this mission. Payload processing and compression functions differently from the previous two calculations, since the compression cost is involved, which affects the raw input being processed. From the 5 KB/s of raw input, we assume a compression cost of 5,000 cycles/byte (k_{comp}).

$$C_{comp} = B_{pl,raw} \cdot k_{comp} = 25,000,000 \frac{cycles}{s} \quad (10)$$

$$U_{comp} = \frac{C_{comp}}{P_{CPU}} = 25\% \quad (11)$$

25% is a lot of the CPU to be taken up by the payload, but it is still under a reasonable quantity and it is expected that this would be the greatest draw of power for the spacecraft. Fault Detection, Isolation, Recover (FDIR) autonomy has a number of factors we assume given the spacecraft constraints. The number of telemetry channels that we monitor for faults is given as N_{ch} . The frequency of FDIR checks in Hz is given as f_{FDIR} . The cost of the CPU per check per channel is given as k_{chk} . Because the need for fast checks is not vastly necessary, f_{FDIR} is assumed to be 1 check per second. The more channels we have the more efficient our processing can be, so 300 channels are assigned. Because of this larger number, the cost is set at 600 checks. The CPU usage is calculated using the following equations:

$$C_{FDIR} = N_{ch} \cdot f_{FDIR} \cdot k_{chk} = 180,000 \frac{cycles}{s} \quad (12)$$

$$U_{FDIR} = \frac{C_{FDIR}}{P_{CPU}} = 0.15\% \quad (13)$$

The need for fault detection is important giving the proximity to long term radiation, that being said, it is still true that FDIR does not require much power. For file systems and storage management, this need for power is even less than expected. With a file system cost of 2,000 cycles (k_{fs}), The Housekeeping and Payload stream storage is a function of the growth factor and the downlink speed:

$$R_{HK,eff} = 15,000 \cdot F = 23.76 \frac{KB}{s} \quad (14)$$

$$R_{pl,eff} = 5,000 \cdot F = 7.920 \frac{KB}{s} \quad (15)$$

$$C_{FS} = (R_{HK,eff} + R_{pl,eff})k_{fs} = 63,360 \frac{cycles}{s} \quad (16)$$

$$U_{FS} = \frac{C_{FS}}{P_{CPU}} = 0.0636\% \quad (17)$$

Given all of the power uses, we compute the total CPU load to be 36.158% of the overall CPU power available. this gives a margin of 63.842%, which is appropriate for a spacecraft as volatile as this one.

D. Bus Throughput Sizing

To ensure the onboard data isn't oversaturated, we scale the bus to be able to handle the worst case use of the data, while allowing a threshold greater than 50%. The utilization of the bus is calculated by:

$$U_{bus} = \frac{\sum_i R_i}{R_{bus,max}} \quad (18)$$

The first term, which is the worst case traffic estimate, is simply the sum of bus and payload, plus margin of error. Previously the total speed is defined as 20 KB/s. With the margin, we assume a worst case of 23 KB/s. The maximum speed of the bus is actually far higher than what we are expected to use, at 300 KB/s. Adjusting for worst case, we have a conservative throughput assumption of ($\eta = 0.5$). With this, we can solve for the equation and find the utilization to be:

$$U_{bus} = \frac{23}{150} = 0.1533 \quad (19)$$

With an estimated 15.33% usage, it is clear we are well within the minimum 50% threshold, and we have ample headroom for issues that may arise from timing errors or burst traffic.

E. Design Tables

Data Source	Rate	Duty Cycle	Volume/day
Housekeeping telemetry	2 kbps	50%	3.6 MB
Payload science data (avg/peak)	5 kbps	100 %	18 MB
Event logs / FDIR	1 kbps	10%	0.36 MB
Total			21.96 MB

Table 4 Preliminary OBDH Data Budget (fill in)

Resource	Selected / Required
CPU performance (MIPS / MHz / cores)	400 MHz
RAM (GB)	16
Non-volatile memory (GB)	1
Mass memory (TB)	3.845
Main bus type + rate	SP0-S, 400 MHz
Estimated bus utilization (%)	15.33%
Time accuracy / synchronization method	<1ms, uplink with DSAC

Table 5 Preliminary OBDH Resource Sizing (fill in)

V. Software, RTOS, Autonomy, and FDIR

A. Software Tasks and Scheduling

In order for a spacecraft with this precise of a mission to fly, organizing how the spacecraft will operate the functions needed for the completion of the mission is necessary. One key piece of technology that will allow for this is a Real Time Operating System (RTOS).

The purpose of the RTOS is to preform all of the automation that required for this mission. This includes orientation and aiming during the Jupiter and Solar Oberth, position correction during cruises between points of interest, determining and fixing any erroneous hardware, and checking if position is correct during the imaging phase while maintaining to say within the focal point.

Another benefit of the RTOS is that it allows for easier automation of task priorities. In the case of this mission, the main priority will always be to keep the spacecraft functional and able to receive commands. This allows for any real time adjustments that were not accounted for during the design phase. What follows depends on the mode the craft is in. For the nominal phases, when the craft is traveling between two points of interest, the next top priority is the positional and rotational determination of the craft, followed by automatic dynamic correction of the spacecraft. For the Oberths, the main priority switches from determination of the position of the craft to making sure the craft is always pointed in the correct direction and that propulsion subsystem is active. Especially in the case of the Sun, where pointing the sail and heat shield is paramount to the completion of the mission within the time frame and the protection of the craft respectively. For the imaging mode, the focus shifts to making sure the craft is positioned in the right spot and also that it is getting good images of the subject planet. For the post mission phase, the priority now just focuses on the health and other auxiliary scientific missions, without worrying too much about propulsion or image capturing.

Below are some details about some of the tasks that will be automated in detail:

Table 6 Software Task Schedule

Task Name	Period / Deadline	Priority	CPU Time	CPU Util.
COM Interface Driver (RX/TX)	Interrupt-driven (<5 ms)	High	50 μ s	5%
Telecommand Processing	Event-driven (<50 ms)	High	200 μ s	3%
Telemetry Packetization	1 Hz	High	1 ms	1%
High-Rate Downlink Streaming (8 kbps)	100 ms cycle	High	2 ms	2%
Downlink Window Scheduler	1 Hz	Medium	200 μ s	<1%
Time Synchronization Update	1 Hz	Medium	100 μ s	<1%
COM Health Monitoring / FDIR	10 Hz	Med-High	300 μ s	3%
File Queue / Buffer Mgmt.	Background	Low	2 ms	5%
Attitude/Navigation support computations.	Event-driven (about 100 ms)	High	5 ms	5%
Payload Image Processing	During Imaging	High	10 μ s	25%

B. Autonomy

When it comes to our ADCS subsystem, this subsystem requires a certain unique level of autonomy. This is because we are dealing with more than 550 AU, and because of this, we have to consider multi-day light-time delays, very narrow antenna beamwidth, and micro-level pointing requirements. Taking these main points into consideration, we know that autonomy is not optional but rather mission-enabling. The concept here for us is high autonomy at the onboard decision-making level. This is where the star tracker itself feeds directly into the ADCS control loop, which can allow the spacecraft to switch sensors, adjust pointing based on performance, and conserve some power. The characteristics here would be power-monitored operations, sensor fusion (tracker, gyro, and sun sensors), FDIR, and closed-loop attitude control. Some examples of where autonomy is required are:

Example 1: Time-Critical Pointing/Power-Aware Operation. A scenario where Time-Critical Pointing needs to be corrected. In this case, if the SGL requires micro-radian pointing accuracy or high-gain antenna alignment with Earth, it would be up to the star tracker to detect the drift, thermal distortion, and maybe saturation of reaction wheels. This then tells the spacecraft to immediately either adjust attitude, de-saturate wheels (thrusters), or reacquire its target. At high speeds, if our power drops, then our star tracker needs to reduce its frame rates, lower image processing frequency, and, if necessary, go into some sort of "standby" mode during its cruise. This would require onboard decision logic.

Example 2: Autonomous Target Reacquisition: In this case, our Solar Gravitational Lens requires precise alignment with distant stars, maintaining a clear focal line position, and making adjustments when certain drift happens. If our star field begins to shift in this case, we need to update our pointing, correct antenna alignment, and all the while having autonomous recalculations of our four-dimensional number systems. This cannot be micromanaged via ground control.

C. FDIR

The Failure Detection, Isolation, and Recovery (FDIR) architecture for the Solar Gravitational Lens mission is built around the reality that the spacecraft must operate on its own for decades in a harsh radiation environment with extremely long communication delays. Once the spacecraft reaches deep space, waiting for ground intervention is not practical, so faults must be detected and handled autonomously. A dedicated hardware watchdog will continuously monitor the active on-board computer (OBC) and will force a reset if expected heartbeat signals are not received, protecting against processor freezes, latch-up events, or radiation-induced corruption. On the software side, additional watchdog functions within the RTOS monitor task timing, processor loading, and subsystem telemetry to identify abnormal behavior early.

VI. Redundancy, Reliability, and Environmental Constraints

A. Redundancy Concept

The Solar Gravitational Lens (SGL) mission will require the spacecraft to operate autonomously for several decades, with a limited opportunity for Earth-based ground intervention. This is because of the extreme communication latency

we will encounter beyond 650 AU. Therefore, the OBDH subsystem must make sure it is highly reliable while also minimizing continuous power consumption. We will be selecting a cold-redundant architecture for the primary On-Board Computer (OBC) due to the power margin of our mission's constrained end-of-life (EoL) from the RTG and to the need for long-term survivability. Using this configuration, we will have one OBC unit that will operate as the active processor, while an identical redundant unit will remain powered off under nominal conditions. Switching between the primary and backup unit will be controlled by a dedicated hardware watchdog and fault management logic. If the active OBC fails to respond within the timing thresholds that we predefine or experiences any repeated resets and internal faults, the watchdog will trigger an isolation of the failed unit and will power on the redundant OBC. This approach is important so that it prevents unnecessary steady-state power draw being associated with hot redundancy while still protecting against single-point processor failure. Redundancy will be implemented at multiple levels within our OBDH subsystem. The processor module and mass memory will be cold redundant. Any critical support electronics like bus transceivers and power regulation interfaces will either be cross-strapped or designed with fault-tolerant components. Memory protection mechanisms (EDAC) and reset capability will also further reduce the probability of us dealing with permanent failure due to radiation-induced upsets.

B. Thermal Constraints

The OBDH subsystem needs to operate reliably across the extreme thermal environments we will encounter during our SGL mission. This includes the high-flux perihelion phase near 0.14 AU as well as the prolonged deep-space cruise beyond 650 AU. Although the thermal control architecture of our spacecraft will be predominantly passive, the electronic components within the OBDH unit must remain within their allowable operating temperature limits under all mission modes. A typical operational temperature range for spacecraft OBDH units is approximately -20°C to $+50^{\circ}\text{C}$. For the SGL mission, the OBDH electronics will be qualified to at least the industrial temperature range (-40°C to $+80^{\circ}\text{C}$). However, we may also have a slight preference towards the military-grade components (-55°C to $+125^{\circ}\text{C}$) when we are far out into our deep-cruise phase, given how long our mission duration is and the radiation environments we will have to deal with. This selection will provide us with some margin against any localized thermal gradients and also transient conditions during our downlink operations or safe-mode configurations. Any form of convective cooling will not be available to use since our spacecraft will be operating in a vacuum environment. Therefore, any heat generated by the OBDH electronics must be removed exclusively through conductive paths to the spacecraft structure and should be subsequently rejected via dedicated radiators. The OBDH units must be mounted to a thermally controlled structural panel so that we can have an efficient level of conductive heat transfer. During our deep-space cruise, internal heat generated from the RTG and neighboring electronics will help us prevent the OBDH from reaching excessively low temperatures, and the multilayer insulation (MLI) will reduce radiative heat loss from occurring. Our thermal design must also account for localized heating during extended communication sessions, since continuous processor activity and transmitter interface operations will increase internal dissipation.

C. Radiation Effects

The SGL mission exposes the OBDH subsystem to a cumulative radiation environment over more than 35 years of operation. This includes the elevated solar particle flux we will encounter during the perihelion phase and galactic cosmic ray exposure on our long-duration deep-space cruise. As a result, both total ionizing dose (TID) and single event effects (SEE) must be considered in the OBDH design. Total ionizing dose leads to long-term degradation of electronic components, which will result in parameter drift, an increase in power consumption, timing instability, or functional failure. To mitigate these effects, we will need to select rad-tolerant or rad-hard components wherever feasible, and shielding of the spacecraft components will also be provided in our dose budget. A conservative TID tolerance level that is consistent with our deep-space missions will be assumed to ensure that we have solid functionality over the full mission duration with an appropriate margin. Single-event effects are caused primarily by individual high-energy particles. They include single event upsets (SEUs), single event latch-up (SEL), and other destructive or non-destructive anomalies. Since we expect to have SEUs over long durations of our mission, the OBDH architecture will incorporate error detection and correction (EDAC) on volatile and non-volatile memory, periodic memory scrubbing, and a watchdog-based reset capability. In the event that we encounter persistent faults or latch-up conditions, our redundancy management system will enable a switchover to the backup OBC. Radiation may also influence factors like our component selection, memory architecture, and processor choice, since certain commercial components may not provide us with proper SEE immunity. Through making use of radiation-tolerant hardware, protective shielding, EDAC implementation, and autonomous fault recovery mechanisms, we can maintain reliable operation in the harsh

heliocentric environment encountered throughout the SGL mission.

VII. Mass and Power Budget

Table 7 OBDH Power by Mode

Component	Qty	Power (W)	Mode(s)
On-board computer / processor module	2	28	LEOP / nominal / safe
Mass memory / storage module	2	10	nominal / downlink
Data bus transceivers / routers	1	15	all
Payload interface electronics	2	10	payload ops
Watchdog / FDIR hardware	3	2	all
Total OBDH		65	

Table 8 OBDH Mass Budget

Component	Qty	Unit Mass (kg)	Total (kg)
On-board computer / processor module	1	0.35	0.35
Mass memory / storage module	2	0.13	0.26
Data bus hardware (routers, transceivers)	1	4.7	4.7
Harness/connectors + margin	20	~0.1	~2
Total OBDH			~7.3

VIII. Interfaces, Risks, Trades, and Verification

A. Subsystem Data Acquisition and Processing

1. COM Interfaces

The OBDH subsystem interfaces directly with the COM subsystem to support telemetry (TM), telecommand (TC), and high-rate science downlink operations. For TM, the OBDH will time-stamp telemetry packets, collect housekeeping data from other subsystems, and prioritize TM content based on operation mode. For TC handling, the OBDH will store and schedule time-tagged commands, update spacecraft clocks for synchronization, adjust limits and operational modes, authenticate telecommands, and execute commands. Finally, for high-rate downlink generation, the OBDH will packetize science data to be compatible with Pulse-Position Modulation (PPM) and Field Programmable Gate Array (FPGA) decoder architecture.

2. ADCS Interface

The ADCS Subsystem data acquisition and processing starts with the star trackers and reaction wheels. Each star tracker acquires and outputs attitude solutions, multiple quality metrics, and several status updates, including temperature and internal faults. Each Reaction wheel provides telemetry and accepts commands such as wheel speed, voltage/power, wheel temperature/status, and, most importantly, the necessary torque. The key data processing involved is time alignment, attitude estimation, guidance/control, wheel torque and momentum management, and FDIR. This is only possible with proper application and synchronization of all sensors. Constant star tracker updates and the rejection of bad measurements using gyros, body rates, and initial pointing feedback are huge contributors to how key data processing is possible.

3. EPS Interface

The Electrical Power Subsystem (EPS) operates closely to the OBDH subsystem, as the OBDH serves as the controller for managing power modes and monitoring. The EPS is designed to generate, store, and distribute electricity, which is then processed by the OBDH through current and velocity from a Power Conditioning Unit (PCU). The OBDH will also keep track of the RTG output, battery state and depth-of-discharge, temperature, and charge/discharge history for assessment of energy balance and degradation trends. Using this data, the OBDH will be able to choose the optimal modes, ensuring the loads stay within the available power generation margins. This will allow load shedding given high-demand operations during the mission, maintaining positive power balance throughout. If there are any fault conditions such as undervoltage, excessive depth-of-discharge, and other anomalies in the RTG, the OBDH shall initiate safe mode automatically. The OBDH is critical in preserving the life of the EPS, ensuring long-term survivability.

4. Payload Interface

The image data is acquired from a telescope designed in series with the Sun as a gravitational lens. Note that the images are in the form of an Einstein Ring, instead of a traditional full circle that is expected when taking images of planets. This data is sent through an encoder that compresses the image with a 2:1 ratio, reducing computation time and resources. Next, the data is then processed in such a way where it is combined with the previous images to form a cohesive planet image. This reduces the kbps that has to be sent as downlink, which minimizes the mass and power of the antenna. Note that as a result of needing to store the medium-term data, a large storage system on the order of terabytes is required.

5. Thermal/Structures Interface

The OBDH unit will be conductively mounted to a thermally controlled structural panel to provide us with a reliable heat dissipation path from internal electronics to the spacecraft radiator network. Since convective cooling is not available in a vacuum, all of our processor and memory dissipation will be transferred through mechanical interfaces and mounting brackets into the primary structure. Localized thermal shielding and multilayer insulation (MLI) are also used to isolate the OBDH from direct radiative exposure during our perihelion phase and to reduce excessive heat loss during our deep-cruise phase. Harness routing is designed to minimize thermal conduction losses while also avoiding routing near high-dissipation elements such as the laser transmitter and RTG interfaces. This strategy will prevent us from avoiding unwanted thermal coupling and localized hot spots.

B. Design trades

A key architectural trade for the SGL mission is hot versus cold redundancy for the OBDH subsystem. Hot redundancy improves availability by keeping multiple processors powered and synchronized. This enables a near-instantaneous failover with minimal interruption to our spacecraft operations. However, this approach also imposes a continuous power consumption and an increased thermal dissipation, which is undesirable with the gradual RTG degradation over the 35+ year mission duration. Instead, our approach will be to use cold redundancy, which would significantly reduce our steady-state power and thermal load by keeping the backup unit underpowered until needed. This would be at the expense of a brief recovery period during switchover. Since we are dealing with a tight end-of-life power margin and our operational profile being predominantly cruise-based, the cold redundancy approach will provide a more balanced solution for our needs. It will preserve a long-term energy margin and reduce continuous thermal stress on our electronics while still maintaining a fault tolerance through a switchover capability triggered by watchdogs.

Another tradeoff made was the choice of the central bus. The main purpose of the bus is to process the housekeeping data and also act as a backup in case radiation inflicts another processor. That means that they do not need much processing power and therefore can be lightweight. The SP0-S accomplishes this while also being radiation hardened for deep-space missions. The payload subsystem however uses 2 NVIDIA Jetson Nano processors instead as a mass efficient way to compute and store the data acquired from the imaging. These will also be radiation hardened, with the 2nd one acting as a backup in case the 1st one becomes unrecoverable. Lastly, the decision as the wires to be a SpaceWire protocol was chosen because it can handle the bit rate from the payload and also has shown to be reliable over many years without being too affected by radiation. Additionally, these wires have low density which minimizes the mass of the spacecraft.

C. Key Risks and Mitigations

When looking at the 35+ year mission duration, deep-space communication, intense radiation environment and small end of life power margins, the OBDH subsystem has many important risks. These risks can be summarized below:

- **Software bugs, integration problems, and autonomous logic error**

The SC must have the ability to operate autonomously throughout the entire mission with multiple long duration light-time delays. The defects in the guidance software, command execution, or FDIR can lead to data corruption and pointing loss. Multiple strategies can be used to mitigate this. Integration testing throughout each system level and independent code reviews / static analysis. Implementing strict configuration management and validating safe-mode transitions through fault testing should also ensure fewer errors.

- **Memory corruption from radiation exposure, and processor upsets**

Other than general degradation from ionization, the long-term exposure to cosmic rays and solar events have the possibility of causing single-event upsets, single-event latch-ups and memory corruption. This can be mitigated using radiation-hardened components, and error detection and correction on the memory leading to periodic memory scrubbing. A cold-redundant OBC can also be cross-strapped with the main OBC, and current limiting can prevent latch-up damage.

- **Bus overload or timing violations**

In the science phase for imaging and high-rate downlink, the CPU bus could exceed its deterministic limits, causing packet loss, telemetry delays and instability.

Via NASA JPL practices, by maintaining a workload margin within 50% of maximum and a bus utilization within 70% of maximum, these issues can be mitigated. Load shedding via mission mode changes can also prevent these issues with automation from OBDH.

- **Thermal margin shortfall**

During perihelion operations and extended downlink, localized heating may occur, while excessive cooling in deep space may also pose risk. Using thermally resistant materials, multilayer insulation, and thermal modeling of each mission phase should assist in mitigating these issues, with a possibility of thermal heating units.

D. Verification and Testing Plan

A thorough testing campaign will aim to simulate each step of the nearly 30-year journey this OBDH subsystem will experience. Testing will be performed at the modular and system level. Hardware-in-the-loop testing will utilize representative sensors to react to anomalies, interpret time-tagged commands, manage science data, perform surveillance on telemetry, execute telecommands, and run through safety mode logic. Environment testing will allow the OBDH to experience representative high levels of radiation, vibrations, thermal-vacuum extremes, and long-duration operations, all while actuating ADCS components, managing EPS power supply, and sending communications. The FDIR system will be tested on its ability to maintain system memory and detect issues. The testing campaign will seek to learn from historical failures in order to catch issues like the 1998 Mars Climate Orbiter's unit conversion error by having rigorous integrated testing.

IX. Conclusion

The OBDH for the SGL spacecraft has been designed in order to fit the positional and rotational accuracy, the automation required, and the tough thermal and radioactive environmental conditions. Some important design choices were to use a centralized, cold-redundant system where housekeeping processes and minimized science and positioning have a majority of the partitioning. Then, using 4 TB and 400 MHz processors, all data that needs to be handled can be with minimal power and mass cost. Additionally, to control for the sensitivity of these state of the art equipment, an FDIR system is put in place. As a result of satisfying the specs, we reach a total dry mass of 7.4 kg where the peak power usage is 68 W during the imaging phase.

A lot of these calculations however assume that the system will be in peak performance with the stats listed on the website. Most of the time, those are over-exaggerations and things can degrade overtime due to radiation. In fact, radiation is the top risk of the mission. One misplaced bit-flip, even with all of our redundancy mechanisms in place, will kill the whole mission. With 39 years of deep space travel, it's highly likely. For the future of a more detailed design, heavy simulation with radiation testing will have to be done to ensure it can handle the dangers of outside the heliosphere.

References

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